

The Anchored Original Offcritical Assembly

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1 One anchored offcritical assembly of the original tau complex

We construct a single diagram from the original source complexes, with a marked arithmetic image and with its other fibres still attached. Its index set does not require naming an actual zero. In particular, the construction makes sense with no decision about the existence of an offcritical zero. The analytic input consists of the exact original calculations (BK.1)–(BK.24), the finite derived comparison (DP.1)–(DP.8), and the original two-leg maps (DC.17)–(DC.23).

1.1 The source-indexed complex and its canonical directed limit

Keep $A = \mathbb{C}[t]$, $D = -x\partial_x$, the original even Schwartz space V with $\phi(0) = 0$ and $\int \phi = 0$, and the original rapid-at-both-ends space $\mathcal{B}$ of (BK.1). All tensor products in this section are algebraic. The operators and entire functions are

$$\begin{aligned} \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), & \mathcal{M}F(s) &= \int_0^\infty F(x) x^s \frac{dx}{x}, \\ Q &= \mathcal{B}/\Theta V, & g(s) &= 2\xi(s), & \mathcal{M}\Theta\phi &= gH_\phi. \end{aligned} \tag{CAU.1}$$

Here $g = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $H_\phi = 2\pi^{s/2}M_+\phi/(s(s-1)\Gamma(s/2))$ with its original entire continuation. The Fourier transform uses $e^{-2\pi i x \nu}$. Set

$$U = \{s \in \mathbb{C} : 0 < \Re s < 1, \quad \Re s \neq \tfrac{1}{2}\}, \quad \mathcal{P}_U = \{p \in \mathbb{C}[t] : p \text{ monic and every root lies in } U\}. \tag{CAU.2}$$

The constant polynomial 1 belongs to this set. Order it by divisibility. It is directed because pP is a common upper bound of p and P . No condition $p \mid g$ is imposed on this index set.

For every $p \in \mathcal{P}_U$ define the following original-source complex in degrees zero, one, and two:

$$\mathcal{K}_p = [V \xrightarrow{v \mapsto (\Theta v, p(D)v)} \mathcal{B} \oplus V \xrightarrow{(F, \phi) \mapsto p(D)F - \Theta\phi} \mathcal{B}]. \tag{CAU.3}$$

The differential squares to zero because $D\Theta = \Theta D$. For $P = rp$, the transition $\iota_{p,P}$ is the identity on degree zero and has the formulas

$$(F, \phi) \mapsto (F, r(D)\phi), \quad F \mapsto r(D)F \tag{CAU.4}$$

in degrees one and two. Applying both differentials verifies the cochain identity, and $\iota_{p,R}\iota_{p,P} = \iota_{p,R}$ because the polynomial multipliers multiply. Thus (CAU.3)–(CAU.4) are one specified functor from the directed category $\mathcal{P}_U$ to complexes of A -modules, not a set of requested properties.

The original Θ is injective and every nonzero $p(D)$ is injective on $\mathcal{B}$ and V . The exact cohomology maps are

$$H^0 \mathcal{K}_p = 0, \quad H^1 \mathcal{K}_p \xrightarrow{\sim} Q[p], \quad [(F, \phi)] \mapsto [F], \quad H^2 \mathcal{K}_p \xrightarrow{\sim} Q/pQ. \tag{CAU.5}$$

Indeed a cycle has $p(D)F = \Theta\phi$, exactly the torsion condition. The source ϕ is unique. If $F = \Theta v$, injectivity of Θ gives $\phi = p(D)v$, so the cycle is the displayed boundary. Conversely every torsion

class has such a representative and gives a cycle. The degree-two quotient is $\mathcal{B}/(p(D)\mathcal{B} + \Theta V)$, which is exactly Q/pQ . Under (CAU.5), (CAU.4) induces the literal inclusion $Q[p] \subset Q[P]$ in degree one, and $[q] \mapsto [r(D)q]$ in degree two.

Define the actual directed colimit complex

$$\mathcal{K}_U = \operatorname{colim}_{p \in \mathcal{P}_U} \mathcal{K}_p, \quad T_U Q = \bigcup_{p \in \mathcal{P}_U} Q[p]. \quad (\text{CAU.6})$$

Its universal property is explicit: a cochain map $\mathcal{K}_U \rightarrow L$ is uniquely the same as cochain maps $f_p : \mathcal{K}_p \rightarrow L$ satisfying $f_{p\iota_{p,P}} = f_p$. To construct it, represent a colimit vector by $x \in \mathcal{K}_p$ and send it to $f_p(x)$. The common-upper-bound relation makes this independent of the representative, and the cochain assertion follows at that stage. Conversely restriction gives the compatible family. Filtered colimits of vector spaces are exact: an element and any relation involving it occur at finitely many stages, and a common upper stage witnesses the required kernel or image identity. This also proves that cohomology commutes with this colimit.

In fact

$$H^1 \mathcal{K}_U = T_U Q, \quad H^i \mathcal{K}_U = 0 \quad (i \neq 1). \quad (\text{CAU.7})$$

Only degree two remains to prove. The original finite-jet calculation (BK.8)–(BK.11) identifies Q/pQ with $A/(p, j_p g)$ by the Mellin jet. At a root ρ of p of order n , let $m = \operatorname{ord}_\rho g$, allowing $m = 0$. The local factor of this cokernel is $\mathbb{C}[z]/(z^{\min(n,m)})$, with $z = s - \rho$; this records the actual unit of g in the Mellin map, although the ideal alone is generated by this power. For $P = p^L$, the transition on degree two is multiplication by p^{L-1} . At ρ this multiplier has zero order $(L-1)n$, while the target cokernel is killed by $z^{\min(Ln, m)}$. Choose L so that $(L-1)n \geq m$ at every one of the finitely many roots with $m > 0$. Every local component then maps to zero. Components with $m = 0$ were zero already. Thus each degree-two class is killed at a later stage, proving (CAU.7). This argument never assumes that $p(D)$ is onto its original test space.

1.2 The exact Mellin identification, including every actual zero

Let $\mathcal{N}_U = (\mathcal{O}_{\mathbb{C}}/(g))|_U$. The notation $\Gamma_c(U, \mathcal{N}_U)$ means sheaf sections with compact support in U . The zeros of the nonzero entire function g form a discrete closed set. Each compact subset of U meets that set in finitely many points. A local section at one of these points extends uniquely by zero outside a sufficiently small disc containing no other zero: on the punctured disc g is a unit and the quotient sheaf is zero. These extensions glue, since their restrictions agree on overlaps. Consequently there is the canonical identification

$$\Gamma_c(U, \mathcal{N}_U) = \bigoplus_{\rho \in U: g(\rho)=0} \mathcal{O}_{\mathbb{C}, \rho}/(g). \quad (\text{CAU.8})$$

The direct sum specifies finite support and keeps the full stalk quotient.

The original Mellin map gives a canonical isomorphism

$$\mathfrak{m} : T_U Q \xrightarrow{\sim} \Gamma_c(U, \mathcal{N}_U), \quad [F] \mapsto ([\mathcal{M}F]_\rho)_\rho. \quad (\text{CAU.9})$$

We prove all assertions, rather than presupposing a spectral identification. If $p(D)F = \Theta\phi$, then

$$p(s)\mathcal{M}F(s) = g(s)H_\phi(s). \quad (\text{CAU.10})$$

Away from the roots of p , this says that $\mathcal{M}F$ is locally a multiple of g . Hence the proposed section has finite support in U . Changing F by an original theta boundary changes its Mellin transform by gH , so the map is well-defined and A -linear.

For injectivity, suppose the section in (CAU.9) is zero. At a root ρ of p of order n , write $p = a(z)z^n$ and $g = u(z)z^m$, with $a(0)u(0) \neq 0$, allowing $m = 0$. The vanishing section says that $\mathcal{M}F = gB$ for an analytic germ B if $m > 0$; it says the same when $m = 0$ because g is then

a unit. Equation (CAU.10) and cancellation in the local analytic domain give $H_\phi = pB$. Thus every jet of H_ϕ of order less than the corresponding multiplicity in p is zero. The proved original range identity (BK.8) gives $\phi = p(D)\psi$ for $\psi \in V$. Then $p(D)(F - \Theta\psi) = 0$, and injectivity on $\mathcal{B}$ gives $F = \Theta\psi$. This proves injectivity.

For surjectivity, take a finitely supported section α of (CAU.8), with support Z , and put

$$p_Z(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad m_\rho = \text{ord}_\rho g, \quad k_Z(s) = g(s)/p_Z(s). \quad (\text{CAU.11})$$

The quotient k_Z is entire, and its germ at each $\rho \in Z$ is a unit. The function g , every multiplicity, and this entire quotient are retained. Choose the unique polynomial P of degree less than $\deg p_Z$ whose jets satisfy

$$[P]_\rho = [k_Z^{-1}\alpha_\rho] \quad \text{in } \mathcal{O}_{\mathbb{C},\rho}/(s - \rho)^{m_\rho}. \quad (\text{CAU.12})$$

Existence and uniqueness follow from the polynomial Chinese remainder map: relatively prime factors have Bezout polynomials, whose products construct the orthogonal idempotents; multiplication by these idempotents and summation realizes any prescribed finite set of Taylor polynomials. The kernel is (p_Z) , and degree reduction gives the unique representative.

Use the original vector $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$, for which $H_{P(D)\phi_*} = P$. The Mellin transform of $\Theta(P(D)\phi_*)$ is gP and vanishes to every order specified by p_Z . The exact original range identity for $\mathcal{B}$ in (BK.8) therefore supplies its unique inverse under $p_Z(D)$:

$$F_\alpha = S_{p_Z}\Theta(P(D)\phi_*) \in \mathcal{B}, \quad p_Z(D)F_\alpha = \Theta(P(D)\phi_*), \quad \mathcal{M}F_\alpha = k_Z P. \quad (\text{CAU.13})$$

Here S_{p_Z} is the composite of the original full-plane Euler inverses at the roots with their full multiplicities. Uniqueness makes the composite independent of the order of those factors. Formula (CAU.12) gives $[\mathcal{M}F_\alpha]_\rho = \alpha_\rho$. Away from Z the quotient by p_Z is a unit at every zero of g , so the other stalk classes are zero. This proves surjectivity. The empty support gives the zero class. Any alternative polynomial lift differing by $p_Z R$ changes F_α by $\Theta(R(D)\phi_*)$, so its class in Q is unchanged. Thus (CAU.13) constructs the inverse to (CAU.9) on the original functions and relations.

In particular, for every actual $\rho \in U$ the element 1 of $\mathcal{O}_{\mathbb{C},\rho}/(g)$ has a nonzero preimage in $T_U Q$. Indeed g vanishes there, so $1 \notin (g)$, and (CAU.9) is injective. Every nilpotent coefficient at that zero is carried by the same inverse. This exhaustiveness assertion is a proved map at every actual zero; it is not an assertion that such a ρ exists.

1.3 The anchored sheaf diagram and its retained diagonal and kernel

Sheafify $\mathcal{O}_U \otimes_A$ applied termwise to (CAU.6), and denote the resulting complex by $\mathcal{A}_U$. Its stalks are the corresponding tensor products because tensor products commute with filtered colimits. Every analytic stalk is torsion-free over the PID A , hence flat by the directed-union-of-finite-free-modules proof in (BK.12). Therefore

$$H^1 \mathcal{A}_U = \mathcal{O}_U \otimes_A T_U Q, \quad H^i \mathcal{A}_U = 0 \quad (i \neq 1). \quad (\text{CAU.14})$$

The natural sheaf map induced by Mellin identifies this degree-one sheaf with $\mathcal{N}_U$. To check it exactly, (CAU.8)–(CAU.9) identify $T_U Q$ as an A -module with the direct sum of $A/(t - \rho)^{m_\rho}$. At a point s_0 , tensoring a summand gives $\mathcal{O}_{s_0}/(s - \rho)^{m_\rho}$ through the original map $a \otimes [P(t)] \mapsto [aP(s)]$. It is zero if $s_0 \neq \rho$, and is the full quotient if $s_0 = \rho$. The Mellin identification retains the unit k_Z used in (CAU.12)–(CAU.13), so this is the actual germ of the map to $\mathcal{O}_{s_0}/(g)$. It is an isomorphism at every stalk and hence an isomorphism of sheaves.

Use the original two-leg complex, with $t = D$ on V_+ and $t = 1 - D$ on V_- , and its analytic comparison:

$$C_\tau = [\mathcal{V}_+ \oplus \mathcal{V}_- \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \widehat{\psi})} \mathcal{B}], \quad C_g = [\mathcal{O}_U \xrightarrow{g} \mathcal{O}_U]. \quad (\text{CAU.15})$$

Their terms are in degrees zero and one and are restricted to U . The comparison $c : C_\tau \rightarrow C_g$ sends (ϕ, ψ) to $H_{\phi - \widehat{\psi}}$ and sends F to $\mathcal{M}F$; on balanced tensors these formulas are multiplied by their analytic coefficient. The identity in (CAU.1) proves it is a cochain map.

There is a cochain map from every $\mathcal{K}_p$ to the original unbalanced joint complex, compatible with (CAU.4), given by

$$v \mapsto \left(\frac{1}{2}v, -\frac{1}{2}\widehat{v}\right), \quad (F, \phi) \mapsto F, \quad F \text{ in degree two} \mapsto 0. \quad (\text{CAU.16})$$

The differential of the first image is exactly Θv because Fourier squares to the identity on even functions. The half factors in (CAU.16) are part of this displayed map; no amplitude is discarded. The map is compatible with every transition because their first degree-one coordinate is unchanged. Taking the colimit and balancing therefore gives $a : \mathcal{A}_U \rightarrow C_\tau$.

Write $C_{\ker} = \ker c$. As proved by the explicit lifts in (DC.20)–(DC.22), there is a short exact sequence of the original complexes $0 \rightarrow C_{\ker} \rightarrow C_\tau \xrightarrow{c} C_g \rightarrow 0$. The single anchored diagram we use is

$$\mathfrak{T}_U^{\text{coh}} := \left(\mathcal{A}_U \xrightarrow{a} C_\tau \xrightarrow{c} C_g, \quad C_{\ker} \hookrightarrow C_\tau, \quad \{\mathcal{O}_U \otimes_A \mathcal{K}_p \rightarrow \mathcal{A}_U\}_{p \in \mathcal{P}_U} \right). \quad (\text{CAU.17})$$

The tensor notation in the final family means its sheafification. This is a representation of the complex-linear path category of the following fully specified quiver with relations: vertices A, C, G, R and one vertex P_p for every $p \in \mathcal{P}_U$; arrows $P_p \rightarrow P_p$ for $p \mid P$, arrows $P_p \rightarrow A$, and $A \rightarrow C \rightarrow G, R \rightarrow C$. Relations identify composites of divisibility arrows, require $P_p \rightarrow A$ to be compatible with them, and require $R \rightarrow C \rightarrow G = 0$. The representation assigns respectively $\mathcal{A}_U, C_\tau, C_g, C_{\ker}$ and the displayed balanced source complexes, with precisely the maps above. The quiver is a set-sized category, since $\mathcal{P}_U$ is a set. Its colimit vertex has the proved universal property following (CAU.6). This specifies one object in a diagram category and its actual entries and arrows, rather than an object defined by satisfying a list of conditions.

The composite ca is a quasi-isomorphism. Both its source and target have cohomology only in degree one, and its degree-one map is the isomorphism (CAU.14) to $\mathcal{N}_U$. With the original notation $\mathcal{M} = \mathcal{O}_U \otimes_A Q$, $\beta : \mathcal{M} \rightarrow \mathcal{N}_U$, and $\mathcal{K} = \ker \beta$, its degree-one factorization is

$$\mathcal{N}_U \xrightarrow{j} \mathcal{M} \xrightarrow{\beta} \mathcal{N}_U, \quad \beta j = 1. \quad (\text{CAU.18})$$

It is the canonical original inclusion of the full torsion summand. Indeed each local class in its image is annihilated by some polynomial, whereas every nonzero polynomial is invertible on $\mathcal{K}$ by BK. Thus it has zero $\mathcal{K}$ component, and β is an isomorphism on that torsion summand. This identifies the actual map induced by (CAU.16), not just an abstract summand of equal length.

For clarity the entire middle term is retained in (CAU.17). On the original joint degree-zero amplitude make the reversible change

$$u = \phi - \widehat{\psi}, \quad v = \frac{1}{2}(\phi + \widehat{\psi}), \quad \phi = v + \frac{1}{2}u, \quad \psi = \widehat{v} - \frac{1}{2}\widehat{u}. \quad (\text{CAU.19})$$

Its differential is Θu ; the v component is the full diagonal. Together with the original canonical splitting $\mathcal{M} = \mathcal{N}_U \oplus \mathcal{K}$, it proves

$$C_\tau \simeq \mathcal{V}_+[0] \oplus \mathcal{N}_U[-1] \oplus \mathcal{K}[-1], \quad C_{\ker} \simeq \mathcal{V}_+[0] \oplus \mathcal{K}[-1]. \quad (\text{CAU.20})$$

These are derived identifications of the explicitly retained complexes. The marked arithmetic image of (CAU.17) is $\text{im } H^1(a)$, which by (CAU.18) is $j\mathcal{N}_U$. Its support is therefore exactly the actual zero divisor of g in U , with its scheme multiplicities. The support of the entire middle term is not being substituted for this specified arithmetic support. In particular the nonzero meromorphic kernel and the diagonal are present even at points where this arithmetic image is zero.

1.4 Full finite observations and all tensor jets of the same diagram

For each $p \in \mathcal{P}_U$ let $E_p = A/(p)$ and let j_p denote full Taylor reduction. The original Mellin maps define the following cochain map, with every factor of g retained:

$$\mathcal{K}_p \longrightarrow [\mathcal{O}_U \xrightarrow{(g,p)} \mathcal{O}_U^2 \xrightarrow{(p,-g)} \mathcal{O}_U], \quad v \mapsto H_v, \quad (F, \phi) \mapsto (\mathcal{M}F, H_\phi), \quad F \mapsto \mathcal{M}F. \quad (\text{CAU.21})$$

On balanced tensors it is $\mathcal{O}_U$ -linear. Applying either differential gives the same expression by (CAU.1) and $\mathcal{M}p(D)F = p(s)\mathcal{M}F$. It induces the original maps to the kernel and cokernel of multiplication by $j_p g$ on E_p ; equivalently its analytic degree-one target is $\mathcal{N}_U[p]$. The local conversion between these two presentations, for $p = a(z)z^n$ and $g = u(z)z^m$, is

$$[v] \longmapsto [(g/p)v], \quad z^{\max(n-m,0)}\mathcal{O}/z^n\mathcal{O} \xrightarrow{\sim} z^{\max(m-n,0)}\mathcal{O}/z^m\mathcal{O}, \quad (\text{CAU.22})$$

with the exact multiplier $(u/a)z^{m-n}$ and inverse $(a/u)z^{n-m}$ on those specified powers. This is a well-defined isomorphism because changing a representative by p changes the product by g , and the source kernel condition makes the product holomorphic. It includes zero factors when $m = 0$ and includes every repeated-zero order. Thus every finite factor of the all-polynomial index is computed, including those which observe no actual zero. In fact (CAU.21), after balancing, is a quasi-isomorphism at every p : the degree-one map is (CAU.22), the degree-two map is the same Mellin identification $\mathcal{O}/(p, g)$, and both degree-zero groups vanish. Flatness and (CAU.5) justify these cohomology calculations for the original balanced source, so no chain-level kernel has been assumed zero.

For an actual finite monic divisor $h \mid g$ with roots in U , the finite packet is the prescribed quotient $A_h = A/(h) \simeq \Gamma(U, \mathcal{O}_U/(h))$. It is obtained from the marked image by the reduction $\mathcal{N}_U \rightarrow \mathcal{O}_U/(h)$ and from the original source by $[F] \mapsto j_h \mathcal{M}F$. Retain the entire quotient $k = g/h$; its germ at a root of h is a unit exactly when h contains the full multiplicity of g there. Both degrees of its derived observation are retained by

$$E_h^{\text{free}} = [\mathcal{O}_U \xrightarrow{h} \mathcal{O}_U] \quad \text{in degrees } -1, 0, \quad - \otimes_{\mathcal{O}_U} E_h^{\text{free}}. \quad (\text{CAU.23})$$

This is a tensor operation on every vertex and arrow of (CAU.17), so it is an observation of the single diagram. On its analytic vertex C_g the total complex in degrees $-1, 0, 1$ is

$$\mathcal{O}_U \xrightarrow{a \mapsto (ha, ga)} \mathcal{O}_U^2 \xrightarrow{(b, c) \mapsto gb - hc} \mathcal{O}_U. \quad (\text{CAU.24})$$

It maps to $[\mathcal{O}_U/(h) \xrightarrow{0} \mathcal{O}_U/(h)]$ in degrees zero and one by $(b, c) \mapsto [b]$ and $d \mapsto [d]$. This is a quasi-isomorphism because it tensors the free complex C_g with the quasi-isomorphism $E_h^{\text{free}} \rightarrow \mathcal{O}_U/(h)$; alternatively its degree-zero cycles are exactly (b, kb) , modulo (ha, ga) , and its degree-one quotient is $\mathcal{O}/(g, h) = \mathcal{O}/(h)$. Under the map to the torsion in $\mathcal{N}_U$ its degree-zero class is $[b] \mapsto [kb]$. The multiplier $k = g/h$ is not dropped.

On the full joint vertex the same observation also retains its diagonal. By (BK.8)–(BK.9), $\mathcal{V}_+/h\mathcal{V}_+ \simeq \mathcal{O}_U/(h)$ through H , and h is injective on $\mathcal{V}_+$. Using (CAU.20) and the contractibility of $[\mathcal{K} \xrightarrow{h} \mathcal{K}]$ therefore gives

$$H^0(C_\tau \otimes E_h^{\text{free}}) \simeq (\mathcal{O}_U/(h))^2, \quad H^1(C_\tau \otimes E_h^{\text{free}}) \simeq \mathcal{O}_U/(h). \quad (\text{CAU.25})$$

In the original plus/minus jet coordinates the degree-zero comparison is $(\alpha, \beta) \mapsto \alpha - \beta$; its kernel is the full diagonal (α, α) . Thus the finite observation does not confuse the joint source with the one-leg complex.

For $h \mid H$, the quotient transition of these observations is the cochain map $E_H^{\text{free}} \rightarrow E_h^{\text{free}}$ equal to H/h in degree -1 and the identity in degree zero. Its cochain identity is $h(H/h) = H$. All transitions of the total diagram are obtained by tensoring this exact map.

To specify all endpoints of these arrows, first adjoin to the seed diagram a vertex (v, h) with value $D_v \otimes_{\mathcal{O}_U} E_h^{\text{free}}$ for every existing vertex v of value D_v and every admitted h . Adjoin the maps $f \otimes 1$ for each original arrow f , the transition maps $1 \otimes (E_H^{\text{free}} \rightarrow E_h^{\text{free}})$, and their cohomology vertices and induced arrows, including the finite quotient vertices in (CAU.24)–(CAU.25). Their relations are exactly the commuting squares supplied by tensor functoriality and by cohomology. This internal tensor operation over $\mathcal{O}_U$ is an explicitly adjoined observation; it is not identified with the external tensor operation in the next paragraph.

Finally close this represented diagram under finite ordered external tensor products over $\mathbb{C}$, with the specified symmetry maps and Koszul signs. The ambient category is graded by the nonnegative integer r : degree- r values are sheaf complexes on U^r , with the coefficient sheaf equal to the tensor product over $\mathbb{C}$ of the pulled-back coefficient sheaves; degree zero is the point with unit $\mathbb{C}$. Permutations use their actual coordinate permutations on U^r . This is a single canonical enlargement: its vertices are finite words of existing vertices, including the empty word; arrows are generated by tensoring the existing arrows with identities, by permutations, and by the displayed finite-observation maps. Their relations are composition, identity, tensor interchange, and the Koszul permutation relations. The value of a word is the literal external tensor product of its original complexes; its differential is

$$d(x_1 \otimes \cdots \otimes x_r) = \sum_{i=1}^r (-1)^{\sum_{j<i} |x_j|} x_1 \otimes \cdots \otimes dx_i \otimes \cdots \otimes x_r. \quad (\text{CAU.26})$$

This construction has the explicit free symmetric monoidal universal property: a tensor-compatible extension of the original represented diagram must have these values and arrows on every word, and the displayed relations make that extension unique. Pairwise interchange of the two differential terms proves $d^2 = 0$ with these signs. For finite packets $h_1, \dots, h_r$, their tensor observation is exactly

$$A_{h_1} \otimes_{\mathbb{C}} \cdots \otimes_{\mathbb{C}} A_{h_r} = \frac{\mathbb{C}[s_1, \dots, s_r]}{(h_1(s_1), \dots, h_r(s_r))}. \quad (\text{CAU.27})$$

Successive monic division in each variable proves the isomorphism: the monomials with $0 \leq j_i < \deg h_i$ give bases on both sides and the map takes the tensor basis to that basis. At specified actual centres it retains all jets $0 \leq j_i < \text{ord}_{\rho_i} h_i$; when each selected divisor contains the full multiplicity, these are exactly all jets $0 \leq j_i < m_{\rho_i}$. It retains every one-variable multiplier $g(s_i)/h_i(s_i)$ from (CAU.24). The total derived observation is the tensor of the two-term observations, so every degree and every tensor nilpotent survives in its prescribed summand. Tensoring over a field is exact: a basis extends a basis of a subspace and splits every vector-space short exact sequence. This proves the cohomological tensor assertion without an unproved flatness claim about a completed analytic tensor product. The tensor sheaves here are sheaves of modules over the literal tensor product of the pulled-back analytic coefficient sheaves; no formal or analytic completion of the original source is implicit.

1.5 All proper source and support fibres, and the exact actions

The diagram has its original label enlargement as part of the same construction. For every D -invariant subspace $W \subset V$ and every nonempty $S \subset \{+, -\}$, include the original complex $C_{W,S}$ with central term $\mathcal{B}$, plus source W if $+$ $\in S$, minus source $\widehat{W}$ if $-$ $\in S$, and differential $\Theta\phi - \Theta\psi$ on the present sources. Give the two sources the actions D and $1 - D$, respectively. Balancing and restriction to U are the same operations used above. For $W \subset W'$ and $S \subset S'$, inclusion on each present source and the identity on $\mathcal{B}$ is a cochain map; the differentials agree term by term. These maps compose literally and are included as arrows in the represented category of (CAU.17). The map to C_g is the original signed H map in degree zero and Mellin in degree one, so these arrows commute with that comparison.

For the joint mask, the change (CAU.19) has $u, v \in W$ and proves $H^0 C_{W, \{+, -\}} = W$ and $H^1 C_{W, \{+, -\}} = Q_W$, where $Q_W = \mathcal{B}/\Theta W$. Each one-leg mask has zero degree-zero cohomology

and the same Q_W in degree one; on the minus leg the actual differential is $-\Theta\hat{\psi}$. The proper-label transition retains the exact original sequence

$$0 \longrightarrow V/W \xrightarrow{[\phi] \mapsto [\Theta\phi]} Q_W \longrightarrow Q \longrightarrow 0. \quad (\text{CAU.28})$$

Injectivity of Θ and the quotient definitions prove exactness; flatness retains it after balancing. Thus every additional proper-label class and its specified vanishing under transition remain in this single diagram. No field-vector assertion for its larger kernel is substituted for the proved assertion about the full source V .

On a support fibre the total carrier is the original disjoint union with external absence. Every displayed linear map f lifts as

$$\tau \mapsto \tau, \quad (W, S, x) \mapsto (W', S', f(x)), \quad e_{W,S} = (W, S, 0). \quad (\text{CAU.29})$$

For a fixed source and target label its supported-zero fibre is exactly $\{(W, S, x) : f(x) = 0\}$, while only external τ maps to external τ . Composition of these lifts follows from composition of the linear maps. Thus the supported part has the actual Grothendieck category of elements: at a fixed cochain degree and open set its objects are the labelled amplitude elements, and its arrows are exactly the represented maps carrying one element to the other. Forgetting the amplitude gives the label projection on this supported part. The single external τ is adjoined to its carrier outside that projection, with the pointed actions in (CAU.29). It is not identified with a labelled object (W, S, τ) of a category of elements. No zero fibre has been replaced by that external point.

The symmetries and actions are also actual arrows. To type their base maps precisely, the enlargement with these arrows uses the category whose objects are ringed spaces with their specified module complexes and whose morphisms consist of a specified base map and the indicated semilinear cochain map. Composition uses the composite base map and the composite semilinear map. Ordinary arrows of (CAU.17) have the identity base map; reflection and conjugation have their displayed base maps. This is the Grothendieck extension of that represented diagram by the following actual action maps. Dilation $R_a F(x) = F(x/a)$, $a > 0$, sends the label W to $R_a W$; on the plus source and target it acts by R_a , and on the minus source it acts by $aR_{1/a}$. The latter lies in $\widehat{R_a W} = aR_{1/a}\widehat{W}$, by direct Fourier substitution. It commutes with the differentials and Mellin takes it to a^s . It acts on (CAU.3) by R_a in each original coordinate and hence commutes with its polynomial transitions and with (CAU.16).

For the reflection $r(s) = 1 - s$ put $JF(x) = x^{-1}F(1/x)$ and $p^r(s) = (-1)^{\deg p}p(1 - s)$, which is again monic in $\mathcal{P}_U$. The r -semilinear chain map from the (W, S) vertex to $(\widehat{W}, \text{swap } S)$ sends

$$(\phi, \psi) \mapsto (-\psi, -\phi), \quad F \mapsto JF. \quad (\text{CAU.30})$$

The theta identity $J\Theta = \Theta^\wedge$ proves the chain identity with the displayed signs, and its square is the identity. On (CAU.3), writing $\epsilon_p = (-1)^{\deg p}$, the corresponding map to $\mathcal{K}_{p^r}$ is

$$v \mapsto \widehat{v}, \quad (F, \phi) \mapsto (JF, \epsilon_p \widehat{\phi}), \quad F \mapsto \epsilon_p JF. \quad (\text{CAU.31})$$

Indeed $\widehat{p(D)v} = p(1 - D)\widehat{v} = \epsilon_p p^r(D)\widehat{v}$ and $p^r(D)JF = \epsilon_p Jp(D)F$. These equalities verify both differentials. For $P = rp$, one has $r^r = \epsilon_r r(1 - s)$ and $\epsilon_P = \epsilon_r \epsilon_p$, which proves compatibility of (CAU.31) with (CAU.4). Formula (CAU.16) intertwines (CAU.31) with (CAU.30) exactly. In (CAU.19) coordinates reflection acts by $(u, v) \mapsto (\widehat{u}, -\widehat{v})$, retaining the diagonal.

On the adjoined finite observations the required cochain map is $r^* E_h^{\text{free}} \rightarrow E_{h^r}^{\text{free}}$, with multiplier $\epsilon_h = (-1)^{\deg h}$ in degree -1 and multiplier 1 in degree zero; indeed $h^r = \epsilon_h h(1 - s)$. On (CAU.24) this sends its three coordinates to $\epsilon_h r(a)$, $(r(b), \epsilon_h r(c))$, and $r(d)$, respectively. Substitution into both differentials proves the cochain identity. For $h \mid H$, the identity $(H/h)^r = \epsilon_{H/h}(H/h)(1 - s)$ and $\epsilon_H = \epsilon_h \epsilon_{H/h}$ proves compatibility with the quotient transition of the observation complexes. Thus the reflection on every observed vertex is typed by its pullback and retains the signs forced by monic polynomial reflection.

Complex conjugation acts on the original functions and all coefficients, takes $p(s)$ to $\overline{p(\bar{s})}$, and commutes with the differentials because the theta sums, D , and g have the corresponding real structure. It commutes with Fourier on the even source spaces. Together with (CAU.30) it gives the original reflected-conjugate operation $s \mapsto 1 - \bar{s}$ on the packet. Direct substitution proves

$$JR_a J = aR_{1/a}, \quad r R_a = aR_{1/a} r \quad (\text{CAU.32})$$

on the labelled chain maps, with the scalar a acting on every term. These are the actual semidirect relations, not a commuting-action replacement. All analytic actions agree because $a^{1-s} = a(a^{-1})^s$ and $g(1-s) = g(s)$.

At an actual zero ρ of order m_ρ the finite arithmetic action obtained from this single diagram is precisely

$$a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad N_\rho[v] = [(s - \rho)v] \text{ in } \mathcal{O}_\rho/(g). \quad (\text{CAU.33})$$

It is multiplication by the Taylor expansion of a^s modulo the full original ideal, so every coefficient and nilpotent order is specified. The finite residue and its weight-one dual action are the original maps (OCQ.26)–(OCQ.30) applied to these same factors; their source inclusion is (CAU.18), and their tensor extensions are fixed by (CAU.26). Thus (CAU.17), enlarged by its explicitly defined label, action, and tensor arrows, is one anchored original-programme object. Its marked arithmetic support is exactly $\mathcal{N}_U$, and (CAU.13) provides a nonzero original-source lift of every nonzero class of each actual offcritical packet. The construction asserts neither that this support is nonempty nor that it is empty.

1.6 The exact attachment to the single arithmetic tensor algebra

The arithmetic tensor algebra (AG14) is attached to (CAU.17) by an explicit source-level map, with no claim that all operations on the ambient test space preserve its arithmetic subspace. Retain its original definitions

$$\mathcal{B}_U = q^{-1}(T_U Q) \subset \mathcal{B}, \quad C_U = [V \xrightarrow{\Theta} \mathcal{B}_U], \quad \mathfrak{T}_U = \bigoplus_{k \geq 0} C_U^{\otimes \mathbb{C}^k}. \quad (\text{CAU.34})$$

These agree exactly with (AG2) and (AG14), since $[F] \in Q[p]$ means $p(D)F \in \Theta V$. Each term has the original topology as a subspace when a topological observation is taken; the displayed DGA itself uses the algebraic tensors specified in AG.

It is essential not to project all of $\mathcal{K}_p^1$ into $\mathcal{B}_U$: an arbitrary F in that degree need not have a torsion class. The exact connecting complex is the good truncation

$$Z_p = \{(F, \phi) : p(D)F = \Theta \phi\}, \quad \tau_{\leq 1} \mathcal{K}_p = [V \xrightarrow{(\Theta, p(D))} Z_p]. \quad (\text{CAU.35})$$

The map $Z_p \rightarrow q^{-1}(Q[p])$, $(F, \phi) \mapsto F$, is a linear isomorphism: existence of ϕ is the defining condition on its target and uniqueness follows from injectivity of Θ . It intertwines the displayed differential with Θ . The transitions (CAU.4) preserve it and become literal inclusions of these target subspaces. Directed exactness consequently proves the cochain isomorphism and the anchored zigzag

$$\mathcal{K}_U \longleftarrow \operatorname{colim}_p \tau_{\leq 1} \mathcal{K}_p \xrightarrow{\sim} C_U \xrightarrow{\sigma} C_\tau^{\text{orig}}, \quad \sigma^0(v) = (\tfrac{1}{2}v, -\tfrac{1}{2}\hat{v}), \quad \sigma^1(F) = F. \quad (\text{CAU.36})$$

Here C_τ^{orig} is the original unbalanced joint complex; its balancing is (CAU.15). The left arrow is the inclusion of the good truncation into the directed complex and is a quasi-isomorphism by (CAU.7). The middle arrow is the actual cochain isomorphism just proved, since the union of $q^{-1}(Q[p])$ is $\mathcal{B}_U$. The right arrow is injective in both degrees and is a cochain map by (CAU.16).

The two maps from the middle complex into the full tau complex agree literally. Balancing preserves this zigzag and its quasi-isomorphism because all the analytic stalks are flat over A . Thus it is the exact source-level attachment of AG's arithmetic complex to the arithmetic vertex of (CAU.17).

Tensoring the right arrow and taking the direct sum gives one injective unital differential graded algebra map

$$\mathfrak{T}_U \hookrightarrow T(C_\tau^{\text{orig}}) := \bigoplus_{k \geq 0} (C_\tau^{\text{orig}})^{\otimes_{\mathbb{C}} k}, \quad x_1 \otimes \cdots \otimes x_k \longmapsto \sigma(x_1) \otimes \cdots \otimes \sigma(x_k). \quad (\text{CAU.37})$$

Its multiplicativity follows from concatenation, its compatibility with the differential follows term by term from (CAU.26), and its injectivity follows by extending bases over $\mathbb{C}$ in every cochain degree before tensoring. The degree-zero tensor unit maps to the same unit. It has the tensor-algebra universal property: every cochain map from C_U into the underlying complex of a unital DGA extends uniquely to a unital DGA map by multiplying the images of the tensor factors, since the Leibniz identity is exactly the signed formula (CAU.26). In particular (CAU.37) is the unique such extension of the specified original source inclusion σ .

Adjoining (CAU.36)–(CAU.37) as arrows to the same represented diagram identifies its arithmetic tensor vertex with the already constructed single DGA (AG14). All its finite packet lifts in (AG6)–(AG13) have the same Mellin image as (CAU.12)–(CAU.13); equality of their classes follows from the injectivity proved in (CAU.9). The ambient vertex retains the full $\mathcal{B}$ and both original tau legs. Accordingly the original logarithmic-derivative and conormal observations remain attached to their actual ambient domains. No closure of $\mathcal{B}_U$ under those operations is asserted, and no observation is silently retyped as an endomorphism of the arithmetic tensor algebra.